

hirerachy

- Association
- Inheritance

has-a relationship

- Human has-a Heart
- Human - Dependent
- Heart - Dependency
- Car has-a Engine

is-a relationship

- Apple is-a Fruit
- Manager is-a Employee

```
class Date{
int day;
int month;
int year;
}
```

Association

1. Composition
 - It represents tight coupling
2. Aggregation
 - It represents loose coupling

```
class Vehicle{
String company;
String vehNumber;
}
```

```
class Employee{
int id;
String name;
double salary;
```

Employee has-a DateofJoining

Employee has-a Vehicle

```
// Association - Composition
Date doj = new Date();
```

In CPP

In Java

By default Composition

By default Aggregation

```
Vehicle veh; //Association - Aggregation
}
```

```
Employee e = new Employee();
```

System.out.println()

Employee e				
empid	name	salary	doj	veh
0	null	0	300	null
int	String	double	Date	Vehicle

200

day
month
year

300

Inheritance

- is-a relationship exists between two entities we use inheritance
- Manager is-a Employee

```
class Employee{
int empid;
String name;
double salary;
}
```

```
class Manager extends Employee{
double bonus;
}
```

```
class Salesman extends Employee{
int noofsales;
double commission;
}
```

```
Employee e = new Employee();
```

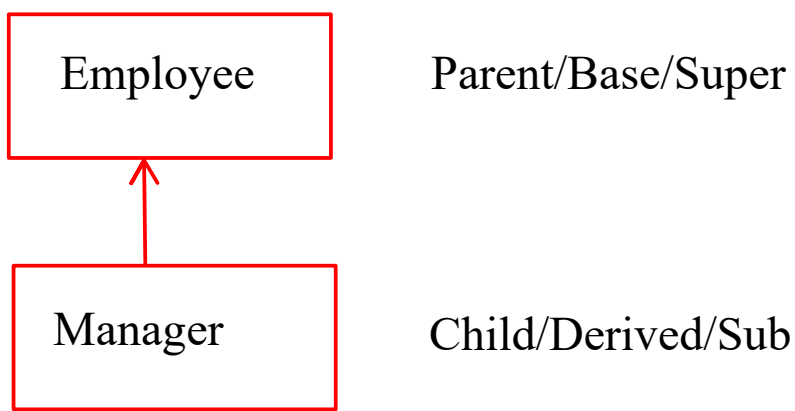
e		
empid	name	salary

```
Manager m = new Manager();
```

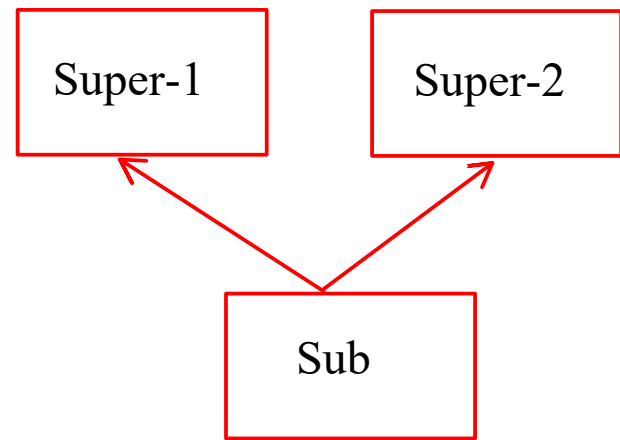
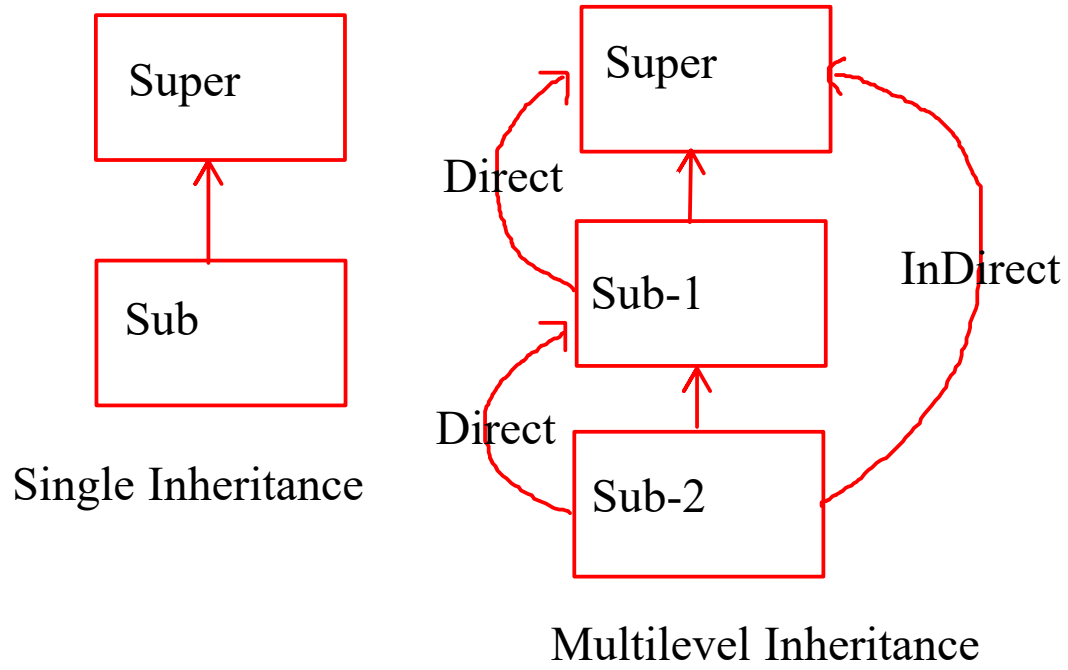
m			
empid	name	salary	bonus

```
Salesman sm = new Salesman();
```

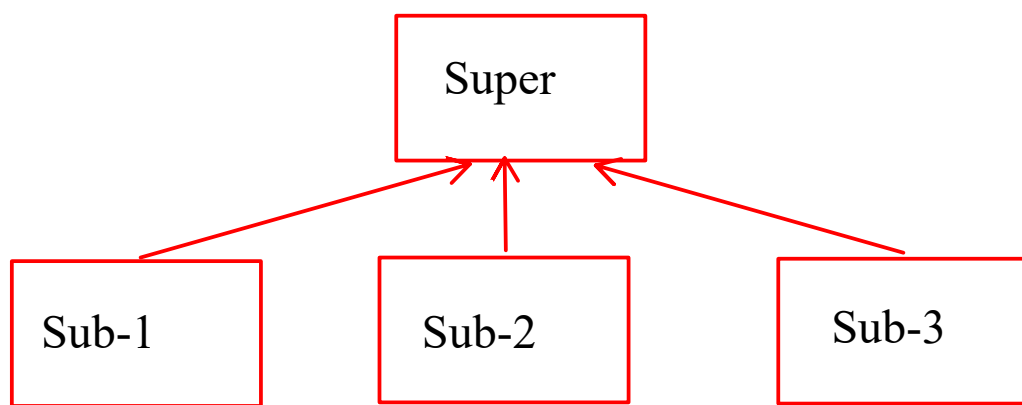
To-Do



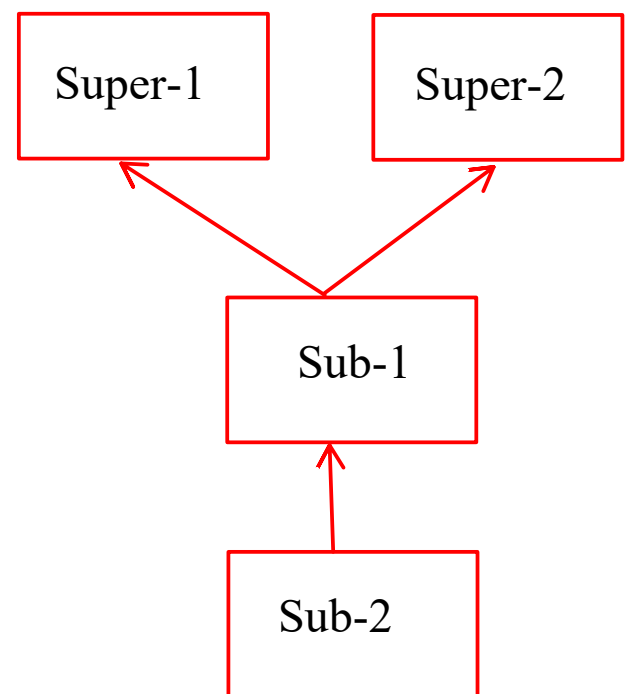
Types of inheritance



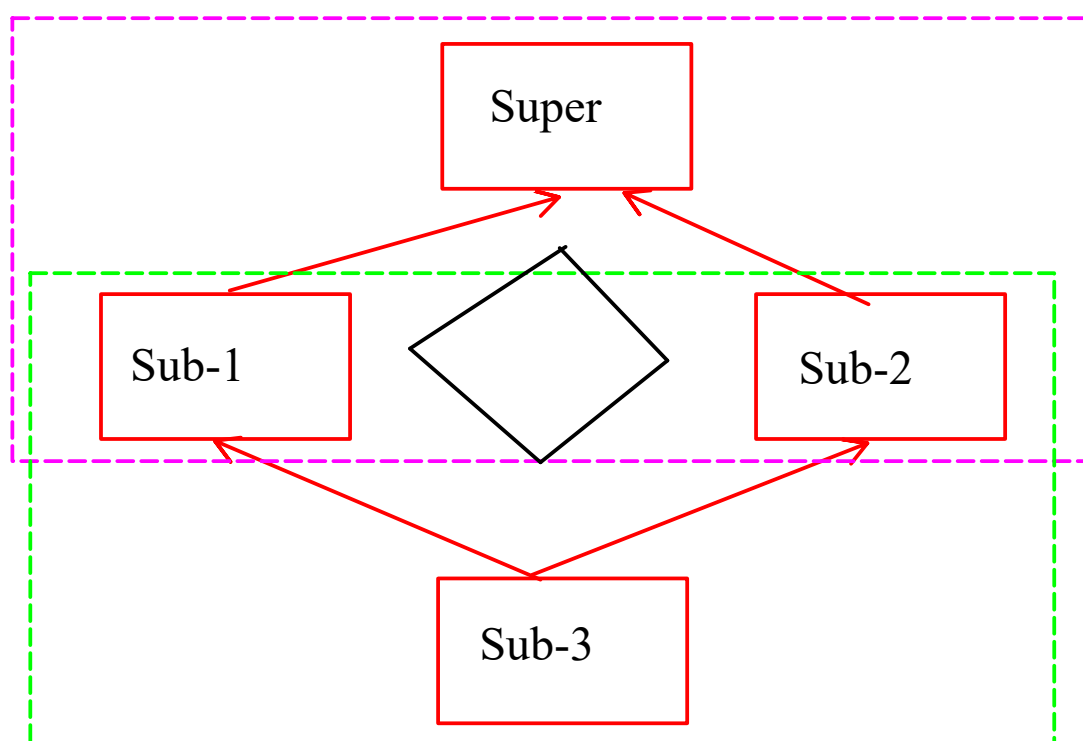
Multiple Inheritance



Hirerachical Inheritance



Mix any two types of inheritance
we get a hybrid inheritance



CPP

In java their is no Diamond Problem
- Reason
- Java does not support multiple
class inheritance